

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

SELF-SUPERVISED LEARNING

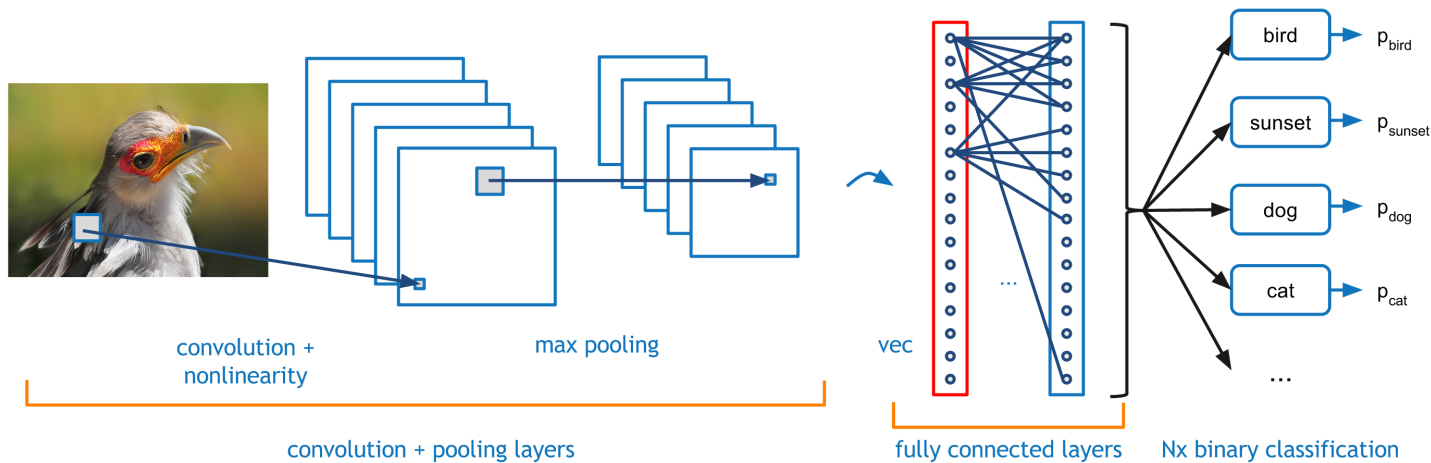
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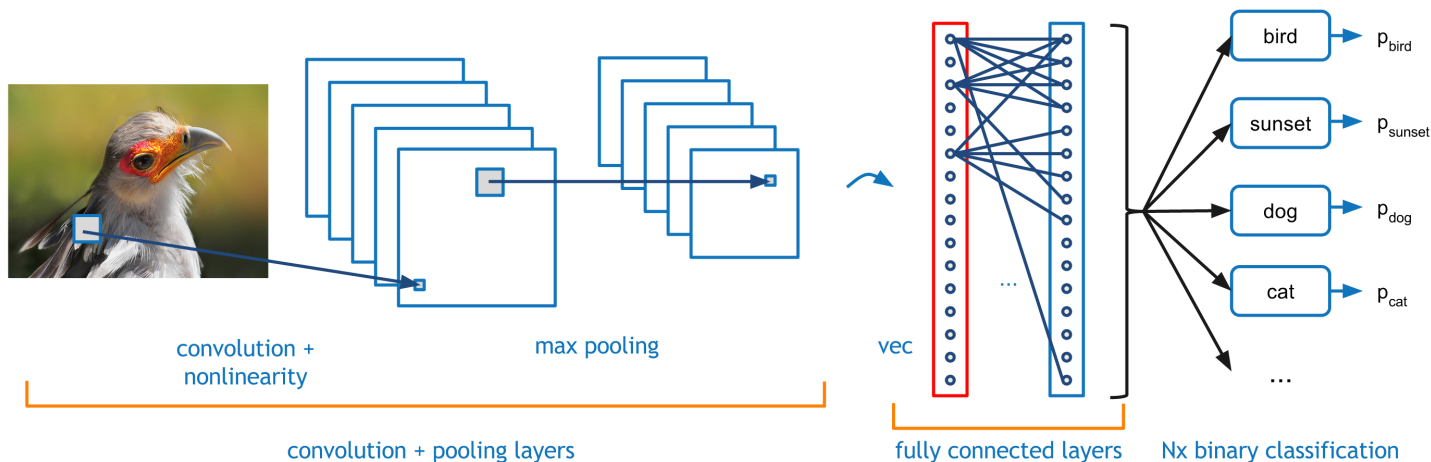
Learning General-Purpose Representations

Successful paradigm: **End-to-end supervised learning** with **lots of labeled data**.



Learning General-Purpose Representations

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Not sustainable!

- Requires a new labeled data set for each new task.
- Instead: general-purpose representations for common data types (images, text).

This is the province of **unsupervised learning**.

Capturing Specific Structure

Precisely define a type of **structure**:

- Clusters
- Subspaces
- Manifolds
- Topics
- Dictionaries

Develop algorithms that uncover this structure in a data set.

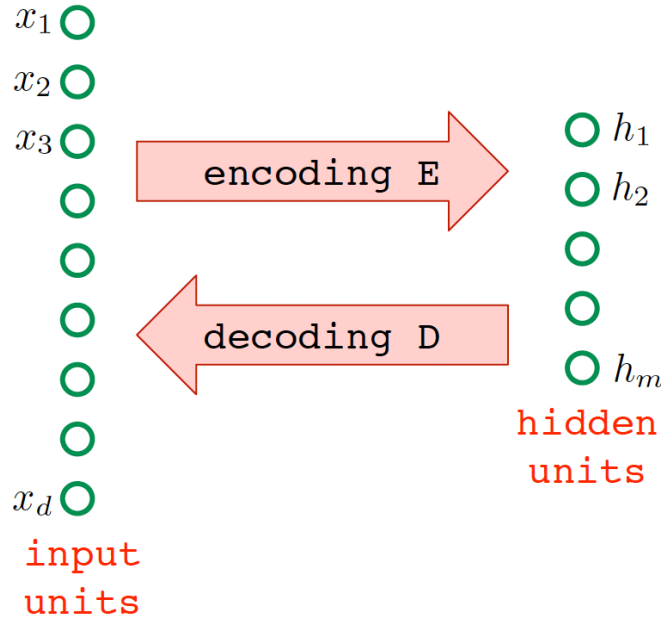
Yields a new representation for each data point x , e.g.,

document $x \rightarrow$ topic distribution $\phi(x)$.

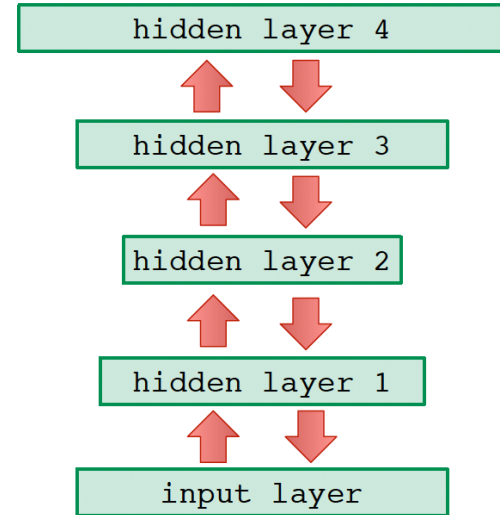
Use this representation in downstream tasks.

More General Autoencoders

Examples: deep belief nets and variational autoencoders.



Stacked autoencoders:



- Loss function: **reconstruction accuracy**. Training can be tough.
- Reconstruction criterion is non-specific in the structure it targets.

Artificial Prediction Tasks

Given an unlabeled data set, create an **artificial prediction task**.

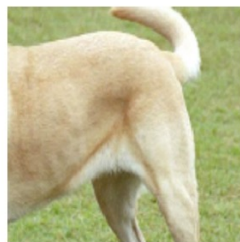
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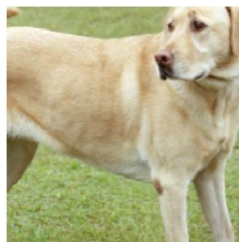
For instance: generate altered versions or **augmentations** of each data point.



(a) Original



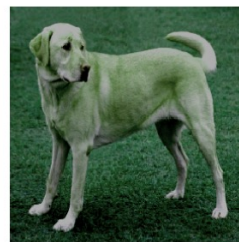
(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



(i) Gaussian blur



(j) Sobel filtering

Chen, Kornblith, Norouzi, Hinton (2020).

Artificial prediction task: Are two augmentations from the same point?

Contrastive Learning

Train a neural net on an artificial prediction problem:

- E.g., For each data point x : generate two variants x' and x'' .
- Create labeled examples:
 - (x', x'') has label $+$
 - (x', z'') has label $-$, where z is some other data point

What is the final **representation** or **embedding**?